Library Management Application

Software Project Management - Week 3 Sample Output

Illustrative Team Alpha
Zhengzhou University
School of Computer and Artificial Intelligence

Lab Practice 03 - Example completed proposal

Abstract

This document demonstrates the expected Week 3 Overleaf output for an example Library Management Application. It includes a project proposal, requirements, three UML diagrams, numbered figures, an equation, cross-references, and cited sources. Students should replace this example with their own project.

1 Project Overview

1.1 What: Project Description

The proposed Library Management Application allows students to search a book catalog, request a loan, view due dates, and return borrowed books. Librarians maintain catalog records and process loans.

1.2 Why: Problem Statement

Manual records make it difficult to check book availability and monitor due dates. A central system can keep book and loan information organized.

1.3 How: Project Plan

1. Gather requirements from students and librarians.
2. Design UML models and a basic interface.
3. Implement and test a prototype.

1.4 Who: Target Users

- Students borrowing books.
- Librarians managing books and loans.

1.5 Expected Outcomes

A prototype with catalog search, loan records, availability checks, and a documented design. Software requirements should be documented before development (Sommerville, 2016).

1.6 Team Members and Roles (Example)

Member	Role	Responsibility
Student A	Project manager	Plan and coordination
Student B	Analyst	Requirements and UML
Student C	Developer	Prototype
Student D	Tester	Test cases

2 Software Requirements

2.1 Functional Requirements

1. FR1: Users can log in.
2. FR2: Students can search the book catalog.
3. FR3: Students can see whether a book is available.
4. FR4: Librarians can issue and record book loans.
5. FR5: Librarians can process returned books.
6. FR6: Students can view their active loans and due dates.

2.2 Non-Functional Requirements

- NFR1: Only authorized users may access personal loan records.
- NFR2: Search results should appear within 3 seconds under the agreed test workload.
- NFR3: Forms and navigation should be clear for first-time users.

2.3 Requirements Table

ID	Requirement	Priority
FR1	Log in	High
FR2	Search catalog	High
FR3	Check availability	High
FR4	Record loan	High
FR5	Process return	High
FR6	View active loans	Medium

3 System Design

The use case model identifies user-facing behavior, while the activity and class models describe workflow and the initial data structure. UML is a standard notation for such models (Fowler, 2003).

3.1 Use Case Diagram

Figure 1 shows the student and librarian actors and their main interactions.

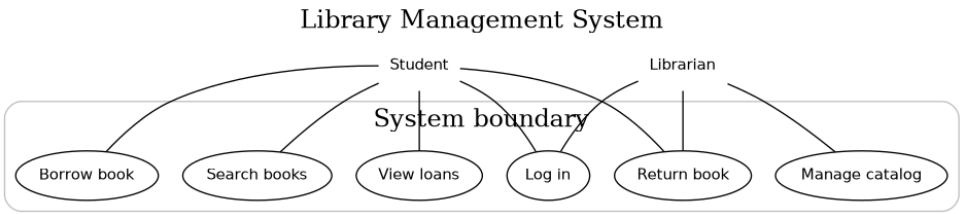


Figure 1: Library application use case diagram

3.2 Activity Diagram

Figure 2 shows the borrowing workflow. A book is issued only when it is available.

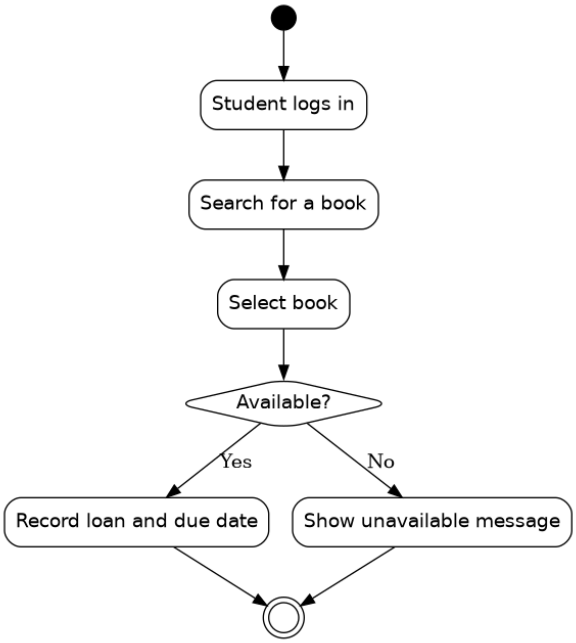


Figure 2: Borrow-book activity diagram

3.3 Initial Class Diagram

Figure 3 presents five initial classes with attributes, methods, and relationships. The design will be refined as requirements become clearer (Pressman and Maxim, 2019).

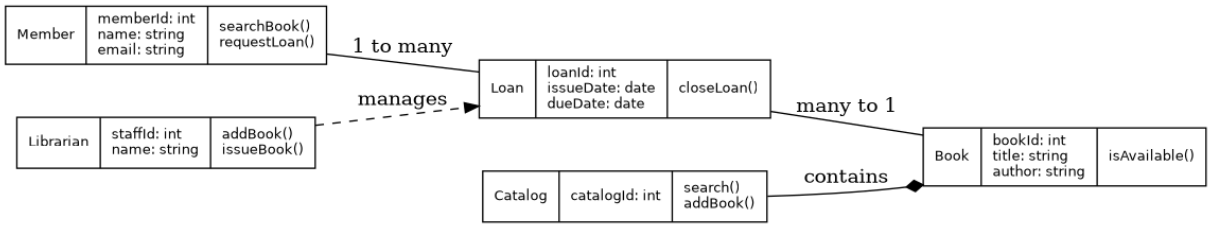


Figure 3: Initial library application class diagram

4 Project Estimation

4.1 Project Completion Percentage

The percentage of planned tasks completed is:

$$P = \frac{C}{T} \times 100 \tag{1}$$

where P is completion percentage, C is completed tasks, and T is total planned tasks ($T > 0$). If 6 of 10 tasks are complete, $P = \frac{6}{10} \times 100 = 60\%$. Equation 1 can be used for weekly progress reporting.

5 Cross-Reference Practice

Figure 1 shows system functions, Figure 2 shows the borrowing process, and Figure 3 shows the initial classes. Equation 1 gives a progress measure. Figure captions, labels, and references are generated automatically in LaTeX.

6 Conclusion

This sample contains the Week 3 outputs. Each team should prepare corresponding requirements, diagrams, calculations, and citations for its own proposed software.

References

Ian Sommerville. *Software Engineering.* 10th ed., Pearson, 2016.

Roger S. Pressman and Bruce R. Maxim. *Software Engineering: A Practitioner's Approach.* 9th ed., McGraw-Hill Education, 2019.

Martin Fowler. *UML Distilled.* 3rd ed., Addison-Wesley, 2003.